Diego Bravo

PhD Candidate | Real-time AI for Medical Imaging

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Research Focus

Real-time AI for medical video understanding, with a focus on procedural auditing via quality indicators and temporal modeling in gastrointestinal endoscopy. Leveraging vision-language models to analyze anatomical bias and spatial reasoning limitations, aiming to improve reliability in uncertainty-aware clinical decision-making. Emphasis on robust, interpretable, and deployable systems for real-world clinical environments, including edge AI and Computer-Aided Diagnosis (CAD).

Education

2023-Present **PhD Candidate in Electrical Engineering**, *Universidad Nacional de Colombia, Bogotá*.
Real-time multimodal AI for medical imaging (25–33 FPS, edge deployment)

2017-2020 **M.Sc. in Biomedical Engineering**, *Universidad Nacional de Colombia, Bogotá*.

2009-2014 **Mechatronic Engineering**, *University: Universidad Militar, Bogotá, Colombia*.

Awards & recognitions

Distinction **MICCAI Society**, *MICCAI Reviewer 2025*, and 2026.

Awards **MICCAI Society**, *MICCAI Conference Registration Grant 2025*, Membership 2026.

Awards **IEEE**, *ISBI Conference*, Student Travel Grant, 2024.

Distinction **Best Paper Award**, *Congreso Nacional de Gastroenterologia*, Asociación Colombiana de Asociaciones del Aparato Digestivo, November 4, 2023. Bogotá, Colombia.

Distinction **Honorable mention (Mención meritoria)**, *Master's degree in Biomedical Engineering.*, presented in School of Medicine at Universidad Nacional de Colombia, November 3, 2020. Bogotá, Colombia.

Distinction **Best Poster Award**, *Congreso Nacional de Gastroenterologia*, Asociación Colombiana de Gastroenterología, January 8, 2020. Bogotá, Colombia.

Distinction **M.Sc. in Biomedical Engineering**, *The best grades average*, University: Universidad Nacional de Colombia, 2017-2018.

Research Experience

2019-Present **Research Assistant**, *at Universidad Nacional, Director: Eduardo Romero, MD. Ph.D.*
- Developed real-time multimodal AI models for gastrointestinal endoscopy analysis
- Designed temporal transformer-based architectures for sequence modeling
- Contributed to publications in IEEE TMI, Nature Scientific Data, MICCAI, and ISBI
- Built deployable systems for edge devices (Jetson Nano)

2021-2022 **Research student**, *at Universidad Nacional de Colombia*, Bogotá, Colombia.

2020-2022 **Research**, Departamento Administrativo Nacional de Estadística (DANE), Colombia.

Work Experience

2023-Present **Research Assistant**, *at Universidad Nacional de Colombia*, Bogotá, Colombia.

2021-2022 **Assistant Professor**, *at Universidad Militar*, Bogotá, Colombia.

2014-2017 **Engineer**, *at Pharmaceutical Laboratory*, Bogotá, Colombia.

2014-2014 **Research Assistant - Undergrade**, *at Universidad Militar*, Bogotá, Colombia.

Publications

Journal Josué Ruano, **Diego Bravo**, Diana Giraldo, et.al., *Spatio-temporal characterization of gastric distensibility in upper endoscopy identifies the presence of Helicobacter pylori*, **IEEE Transactions on Medical Imaging - TMI**, 2026.

Conference **Diego Bravo, et.al.**, *When Vision-Language Models Look but Don't See: Anatomical Bias in Endoscopic Spatial Reasoning*, ISBI2026, London, England, Apr 2026, **In-Press**.

Journal **Diego Bravo,** Juan Frias, Felipe Vera, et.al., *GastroHUN an Endoscopy Dataset of Complete Systematic Screening Protocol for the Stomach*, **Scientific Data - Nature**, Nature Publishing Group UK London, 2025.

Conference **Diego Bravo, et.al.**, *Automated Auditing of Upper Endoscopy Procedure Times: A Temporal Multiclass Analysis*, **Main Conference MICCAI 2025**, Sep 2024.

Conference **Diego Bravo, et.al.**, *Self-Supervised Learning for Multi-Category Endoscopy Classification and Data Quality Evaluation Using Masked Autoencoders*, ISBI2025.

Conference **Diego Bravo,** Sebastian Medina, Josué Ruano, et.al., *Automatic Endoscopy Classification by Fusing Depth Estimations and Image Information*, ISBI2024, May 2024.

Conference Josué Ruano, **Diego Bravo**, Diana Giraldo, et.al., *Estimating Polyp Size From a Single Colonoscopy Image Using a Shape-From-Shading Model*, ISBI2024, May 2024.

Conference Maria Jaramillo, **Diego Bravo**, Josué Ruano, et.al., *Predicting Metaplasia in Upper Gastrointestinal Images from White Light Endoscopy*, ISBI2024, May 2024.

Conference **Diego Bravo, et.al.**, *Automated Anatomical Classification and Quality Assessment of Endoscopy by Temporal-Spatial Analysis*, SIPAIM2023, November 2023.

Conference Josué Ruano, **Diego Bravo**, María Jaramillo, et.al., *Generating synthetic endoscopy videos following a systematic screening protocol*, SIPAIM2023, November 2023.

Conference María Jaramillo, Josué Ruano, **Diego Bravo**, et.al., *s*, Automatic Localization of Pancreatic Tumoral Regions in Whole Sequences of Echoendoscopy Procedures, SIPAIM2023.

Conference **Diego Bravo,** Josué Ruano, María Jaramillo,et.al., *Automatic Classification of Esophagogastroduodenoscopy Sub-Anatomical Regions*, ISBI2023, April 2023.

Abstract Fabian Emura, **Diego Bravo**, Josué Ruano, et.al., *ID: 3526254 A Novel Deep Learning Model to Facilitate Complete Systematic Photodocumentation During Upper GI Endoscopy*, Gastrointestinal Endoscopy, May 2021.

Conference **Diego Bravo,** Josué Ruano, Martín Gómez, Eduardo Romero, *Automatic polyp detection and localization during colonoscopy using convolutional neural networks*, SPIE Medical Imaging 2022, Houston, Texas, United States, February 2020.

Conference **Diego Bravo,** Josué Ruano, Martín Gómez, et. al., *Automatic polyp localization by low level superpixel features*, SIPAIM2019, Medelin, Colombia, October 2019.

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Google Scholar: https://scholar.google.com/citations?user=S47_4IcAAAAJ